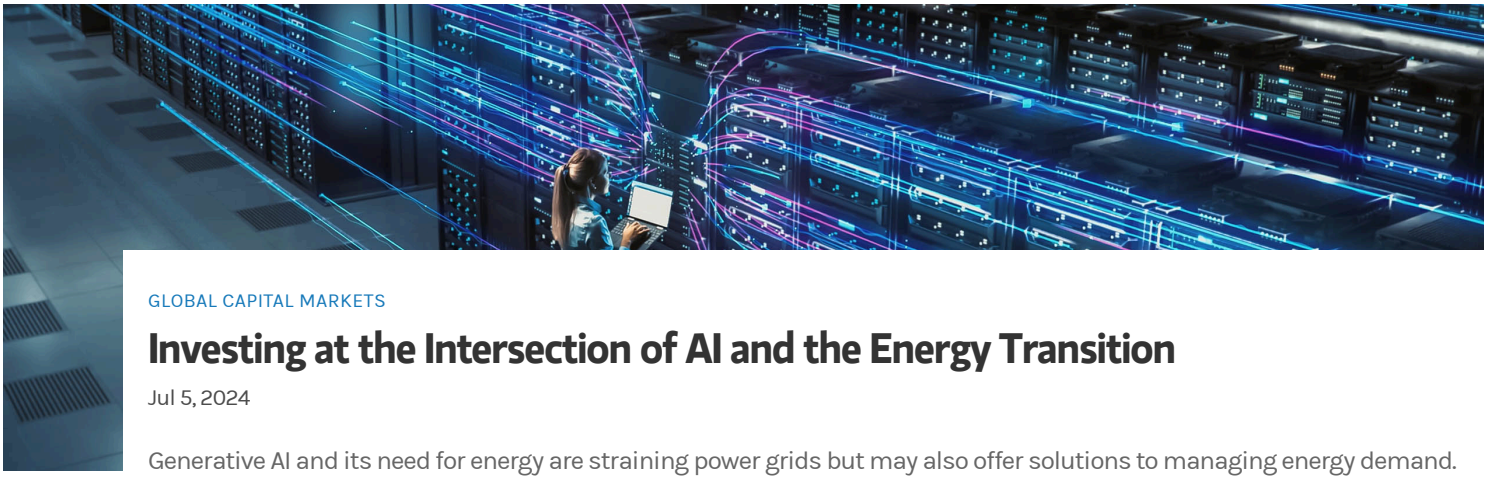

GLOBAL CAPITAL MARKETS

Investing at the Intersection of AI and the Energy Transition

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Generative AI and its need for energy are straining power grids but may also offer solutions to managing energy demand. Investors and corporate leaders are assessing the sustainable innovations that could help to meet this challenge.

Key Takeaways

- The clean energy transition and mass uptake of artificial intelligence (AI) are converging, creating potential investing opportunities.
- Investors are assessing solutions that can address high energy demand and power grid reliability while reducing climate risks.
- Sustainability innovations aim to tackle power transmission limitations, energy storage and greenhouse gas emissions.
- The market for sustainability bonds has reached a new record, and new types of ESG-labeled debt include financing for nuclear energy projects.

Before **generative AI** became a focal point for most companies and investors, the clean energy transition was one of the most compelling and complex business opportunities in global markets. Now, the convergence of these transformative trends has posed a conundrum: On one hand, the massive energy requirements of AI data centers will cause a surge in power demand,

As generative AI and the clean energy transition move toward an inflection point, Morgan Stanley's 2024 Sustainable Finance Summit gathered corporate executives, sustainability leaders and institutional investors to discuss how capital markets, governments and technology can address these critical issues. **Lisa Shalett**, Chief Investment Officer, Wealth Management at Morgan Stanley, posed one version of a key question that was voiced repeatedly at the summit: "How are we going to digitize at the rate and pace that everyone thinks we should, without fundamentally upgrading the grid and power generation capacity around the world?"

To meet the ever-growing need for energy in a sustainable way in the coming years, there are three important areas to consider:

1. Continuing to advance clean energy goals without compromising access to power
2. Investing in sustainable innovations
3. Growing and evolving approaches to sustainable finance

1. Bridges to Renewable Energy and Solving Grid Reliability

Generative AI's power demands could skyrocket 70% annually, and by 2027, it may use as much energy as all of Spain needed in 2022, according to **Morgan Stanley Research**. In addition to AI, the rise of electric vehicles and the world's standard electricity needs, such as heating and cooling, are set to heavily strain global power grids.

With proliferating energy demand, and the variable nature of renewables such as wind and solar, companies and investors are monitoring what mix of energy solutions may be required to fuel the world's power needs. "There are debates about the messiness of the energy transition," said Stephen Byrd, Morgan Stanley's Global Head of Sustainability and Clean Tech Research. "For AI companies, reducing the time needed to get power is the most important commodity. And while most data centers will be clean-energy powered, not all will."

Summit panelists shared the following trends related to the need for more green and consistent power:

- **Bridging "Dirty" and "Clean" Energy:** Natural gas may reduce dependence on more polluting fossil fuels while filling gaps in supply left by wind and solar power, in order to power consistent electricity. Using natural gas may be necessary for the next 30 years or more, some estimated.

- **Solving for Peak Demand Constraints:** A common problem worldwide is that during peak hours, power grids lack enough energy to meet demand. Companies and investors are assessing AI tools that could help utilities forecast energy spikes to adjust their distribution strategies, and battery storage systems that can set aside clean energy during off-peak hours and deploy it when demand is high.

- **Shifting to Energy-Efficient Systems:** Companies and consumers are adopting technologies that can reduce energy consumption, especially in heating, ventilation and air conditioning (HVAC) systems. Energy-efficient heat pumps that use electricity to transfer heat are replacing traditional setups that generate heat through the combustion of fossil fuels

- **Devising Government Incentives to Save Energy:** Government agencies in the U.S. and Europe are motivating consumers to transition to energy-efficient systems at scale. Some are offering incentives or subsidies for systems such as heat pumps or battery storage, while others initiated legislation that would ban fossil-fuel heating in homes.

2. Opportunities in Sustainability Innovations

Summit attendees agreed that the energy transition and provision of reliable power require the adoption of innovations that can scale with the help of capital markets. They highlighted the following emerging technologies and newer mechanisms that could have high impact:

- **Better Transmission Infrastructure:** To meet growing demands for clean electricity, governments will need to expand and upgrade the systems that deliver power over long distances. Currently, these transmission lines heat up when power runs through them, limiting their capacity. Technologies powered by AI are helping grid operators retrofit transmission lines with weather monitors to adjust the carrying capacity in real time during cooler weather, which maximizes the flow. Investors are assessing opportunities to invest in companies that facilitate these types of improvements in transmission infrastructure.

- **Batteries and Fuel Cells:** Lithium-ion batteries store and release energy, while fuel cells use hydrogen to generate electricity. Some see potential in these technologies to help improve power consistency, but challenges include batteries' overheating and fire risk, and the high costs and lack of infrastructure for fuel cells.

- **Methane Offsets:** Scientists and policymakers increasingly recognize the importance of reducing methane emissions, which are responsible for about 30% of the current rise in global temperature[1] and are on track to exceed carbon dioxide emissions by a factor of approximately 30 in the next 20 years.[2] Compared to carbon offsets, methane-offset projects are newer and less widely adopted, and they focus on reducing and removing methane by combusting it into less harmful molecules—often around the sites of landfills or coal mines—or using the gas for purposes like heating water or generating electricity. As with assessing carbon offsets, methane-offset buyers are analyzing methodologies to determine whether methane reductions in projects are genuine and quantifiable.

3. Growth of Sustainable Financing and Green Bonds

The bond market for environmental, social and governance (ESG) projects reached a record in the first quarter of 2024,[3] as green, social, sustainability and sustainability-linked bonds added $272 billion in volume. "The growth of sustainability bonds is one of the mega secular trends that's going to affect capital flows for the foreseeable future," said **Melissa James**, Vice Chairman of **Global Capital Markets** at Morgan Stanley. "Investors are interested in the financial prospects of sustainability projects that align with their sustainable investing goals."

In addition to highlighting the sustainability bond market's growth, panelists spoke about new types of ESG-labeled debt that

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- **Nuclear Energy & Carbon Capture in Green Bonds:** [illegible] starting to [illegible] green bonds [illegible] financing [illegible] projects that extend the life and safe operations of existing nuclear assets, or help convert gas assets for carbon

transport and storage," said **Barbara Calvi**, EMEA Head of Fixed Income ESG Strategy & Research at **Morgan Stanley Investment Management** and **Calvert**. She highlighted the importance of evaluating the merits of new technologies and green financing on an ad hoc basis, to support investments that can credibly contribute to the energy transition.

- **Outcome Bonds:** The World Bank recently issued a new type of sustainability bond called an outcome bond, which finances sustainability projects but makes a portion of the return on the bond contingent on the success of those activities.[4] Different from other types of sustainability bonds, project performance risk is transferred to investors, who might forgo ordinary bond coupons to receive potentially higher coupons based on the success of the underlying sustainability projects. Outcome bonds are funding projects in wildlife conservation, emissions reductions and water purification.

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